



PROJECT (Phase 4): MEDICO (HEALTH CARE SYSTEM)

**SUSTAINABLE DEVELOPMENT GOALS: GOOD HEALTH AND WELL
BEING**

GROUP 3

SECV2113-01 HUMAN COMPUTER INTERACTION

SEMESTER II, SESSION 2023/2024

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Table of contents

1.0 Introduction.....	3
2.0 Digital photos of the pieces of the prototype	4
2.1 Task 1: Review the self-monitoring analysis	6
2.2 Task 2: Book health care services	11
2.3 Task 3: Explore the chatting platform.....	16
3.0 Briefing.....	18
4.0 Testing with users.....	19
Task 1: Review the self-monitoring analysis.....	19
Task 2: Book health care services	20
Task 3: Explore the chatting platform	21
4.1 User 1:.....	22
4.2 User 2:.....	22
4.3 User 3:.....	22
5.0 Observation	23
6.0 Findings.....	25

1.0 Introduction

The Medico app is an improved version of the SEHATi app, and 7 Cups app aimed at enhancing the user experience particularly for elderly users, physical health user and mental health user aligning with the Sustainable Development Goals of Good Health and Well-being. Developing a prototype has provided us with many benefits and drawbacks. In addition to creating the prototype, we can learn about everything, particularly analytics and heuristic interfaces. The development of a model from a sketch to a prototype is not an easy process. As technology advances daily, every bit of knowledge we learn from scratch and throughout this project will be helpful to people.

The usability testing for the Medico app was conducted during the second semester of the 2023/2024 academic session. We utilise the Figma platform, a web-based design tool that allows for creating and testing interactive prototypes. It is accessible to draw the prototype for our application and provide free service to the public. Besides that, Figma is also available in many types of devices like PC, phones or tablets.

The two main tasks chosen for observation from the existing system are the SEHATi app and 7 Cups. The task analysis was conducted to identify and establish requirements for our product which is an application. Three primary tasks were tested during the usability evaluation is track health data efficiently, book medical checkup appointment easily and straightforward, and obtain mental health support in busy life. Three users with different backgrounds were selected for user analysis including individuals with physical health problems, elderly individuals, and individuals with mental health issues. Three users with different backgrounds were selected for user analysis: UTM students, staff and visitors. These three different users made it easier for us to understand how different users approach completing tasks, the limitations they might encounter, and their preferences or expectations for the application's features.

2.0 Digital photos of the pieces of the prototype



Figure 1: Login page

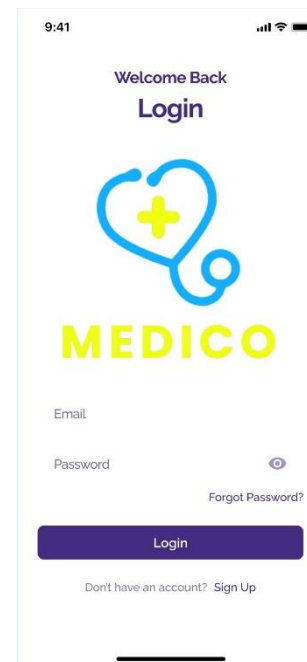


Figure 2: Login page

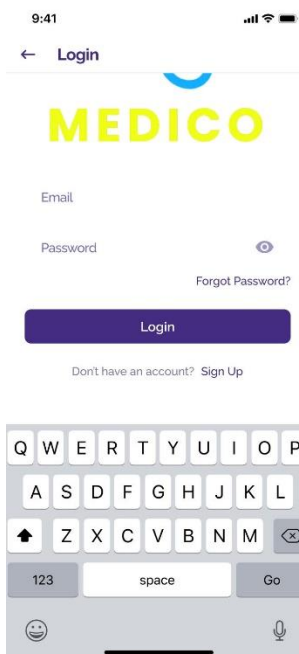


Figure 3: Log in page

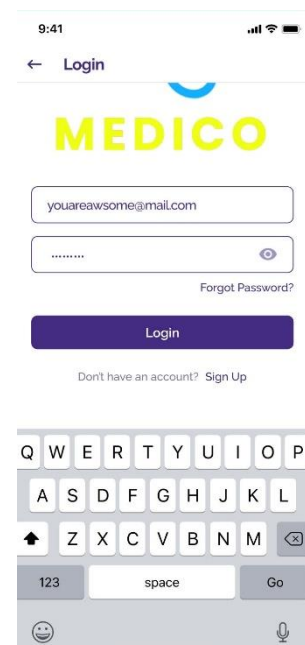


Figure 4: Log in page (Fill in information)

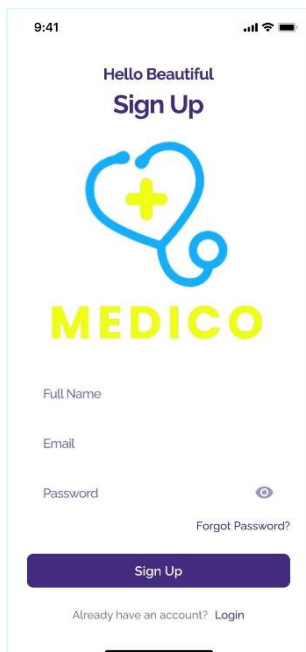


Figure 5: Sign up page

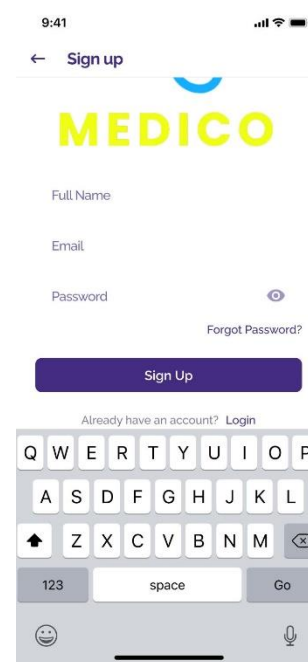


Figure 6: Sign up page(Fill in information)



Figure 7: Sign up page (Fill in information)



Figure 8: Main page

2.1 Task 1: Review the self-monitoring analysis



Figure 9: Self Monitoring page

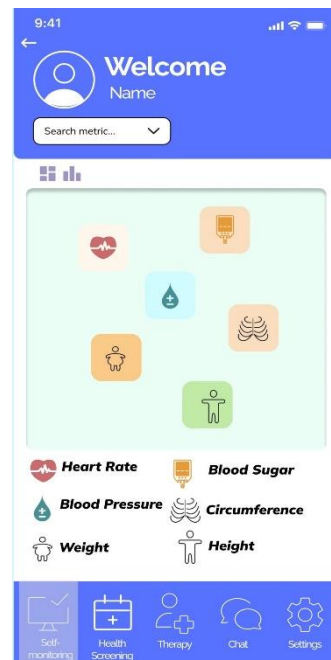


Figure 10: Self Monitoring page



Figure 11: Blood sugar analysis

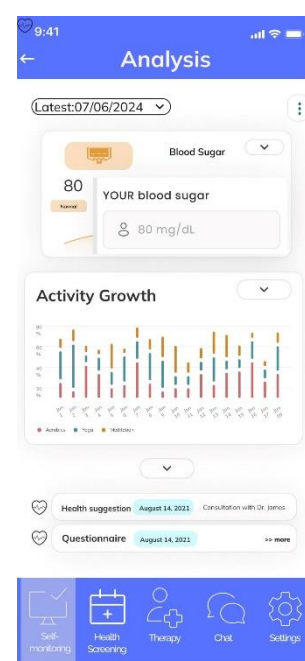


Figure 12: Blood sugar analysis



Figure 13: Heart rate analysis

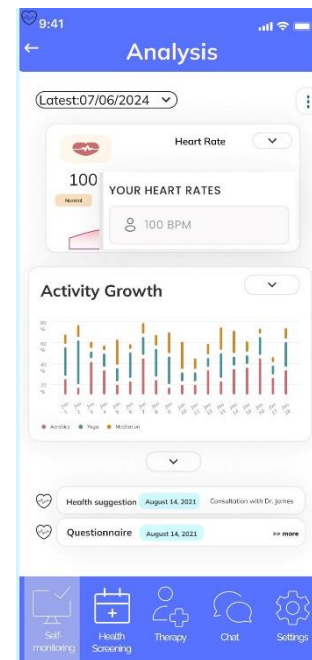


Figure 14: Heart rate analysis



Figure 15: Blood pressure analysis



Figure 16: Blood Pressure analysis



Figure 17: Weight analysis

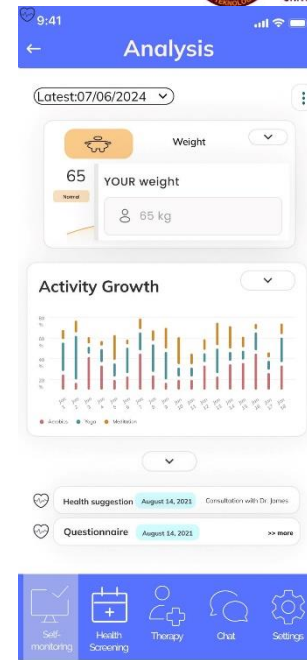


Figure 18: Heart Rate analysis



Figure 19: Height analysis

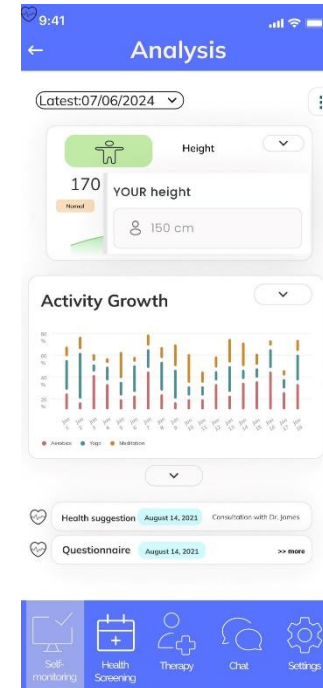


Figure 20: Height analysis



Figure 21: Circumference analysis

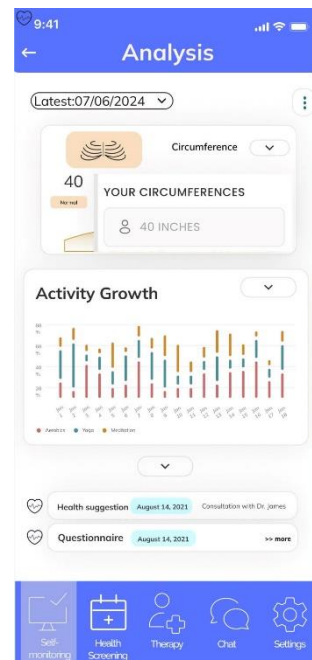


Figure 22: Circumference analysis

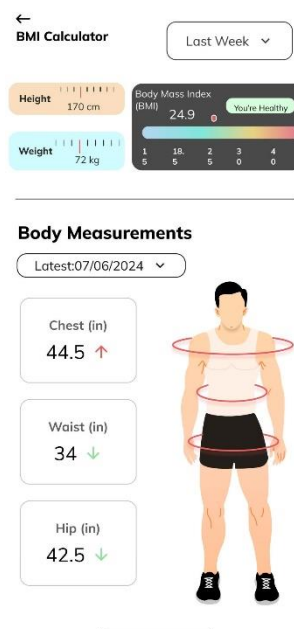


Figure 23: BMI Calculator

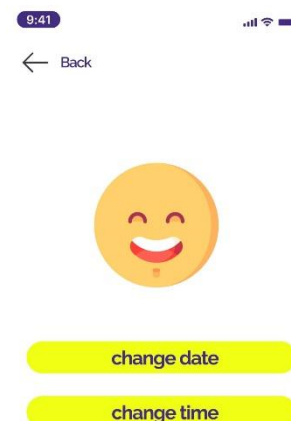


Figure 24: Change date

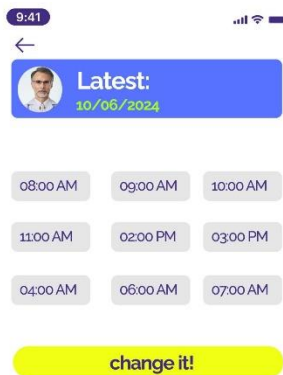


Figure 25: Change Time

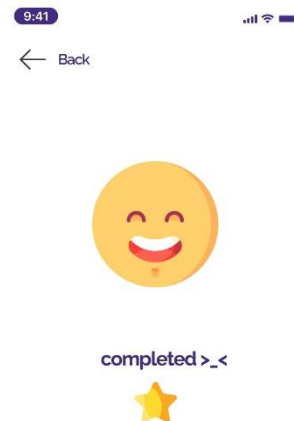


Figure 26: Complete Time and Date



Figure 27: Activity Growth page

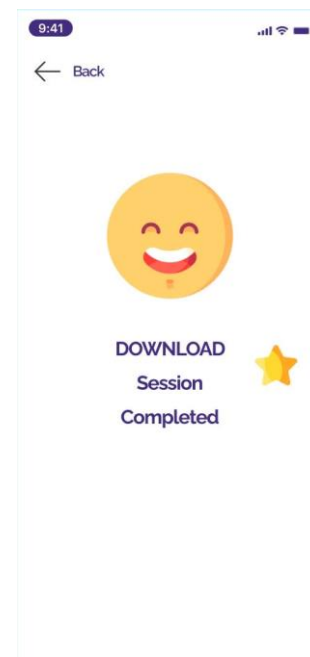


Figure 28: Download Complete page

2.2 Task 2: Book health care services

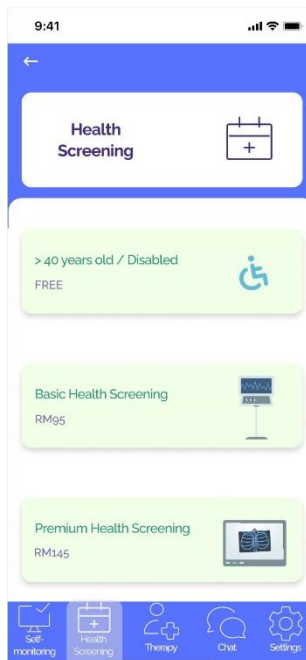


Figure 29: Health screening page



Figure 30: Map page

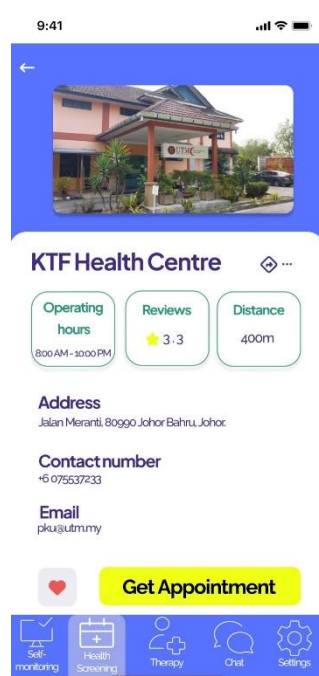


Figure 31: Health Centre information

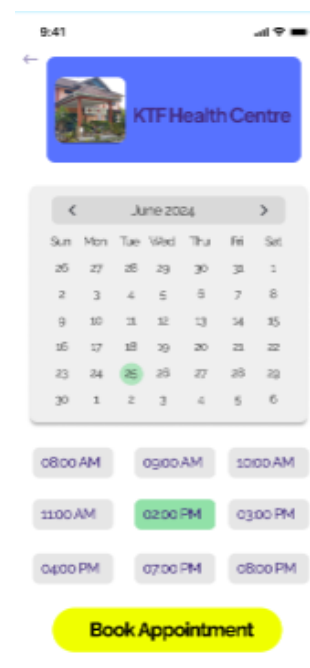


Figure 32: Book appointment page

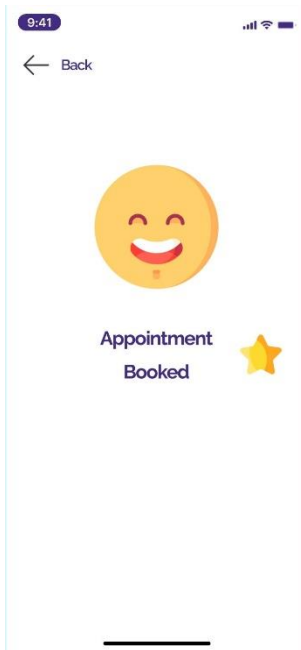


Figure 33: Appointment Booked Done page

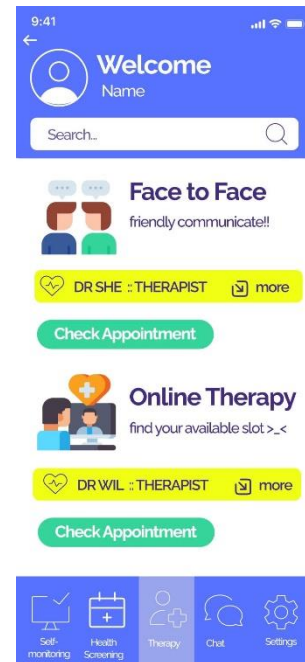


Figure 34: Therapy page



Figure 35: Face to Face Therapy page

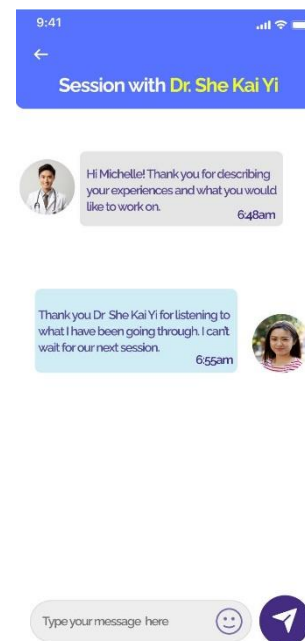
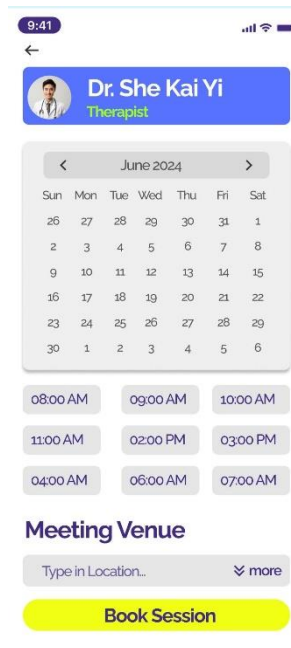
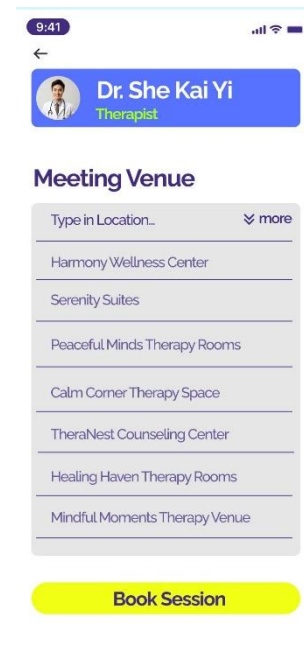


Figure 36: Chat with doctor page



**Figure 37: Book session page
(Choose Time and Date)**



**Figure 38: Book session
page(Choose Location)**

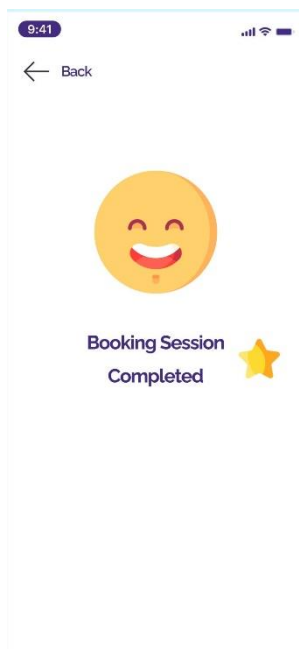


Figure 39: Booking complete page

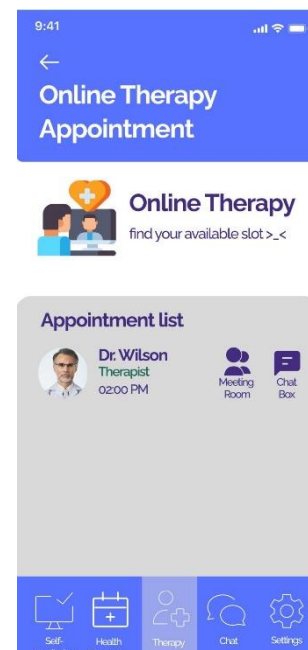


Figure 40: Online therapy page



Figure 41: Doctor profile page



Figure 42: Online meeting with doctor

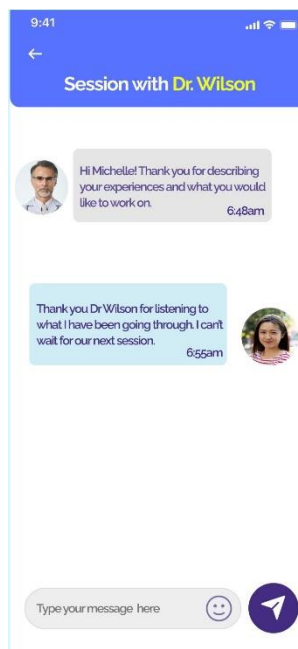


Figure 43: Chatting with doctor page

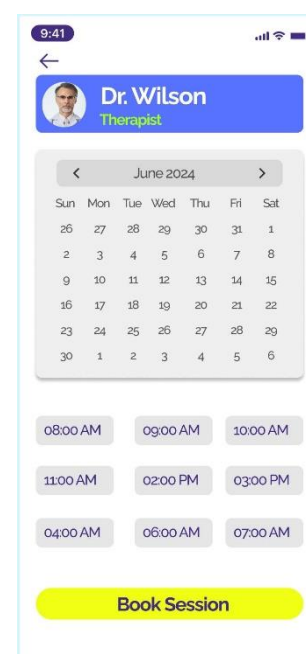


Figure 44: Booking session page(Choose Time and Date)

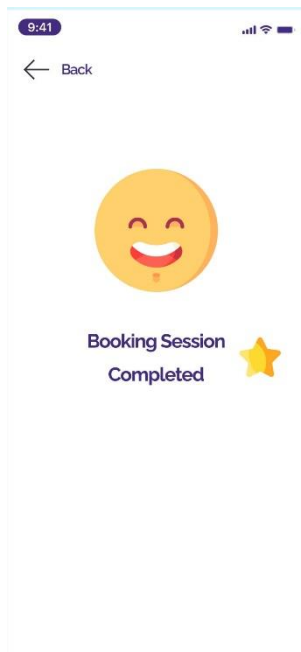


Figure 45: Booking complete page

2.3 Task 3: Explore the chatting platform



Figure 46: Chat page

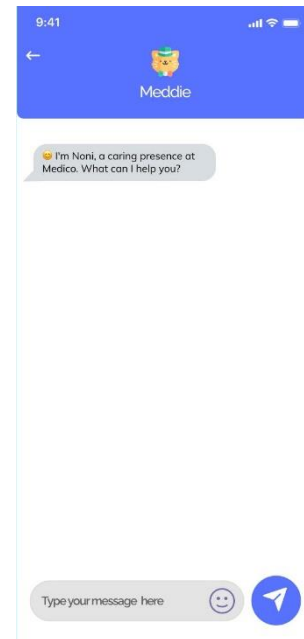


Figure 47: Chat with AI chat

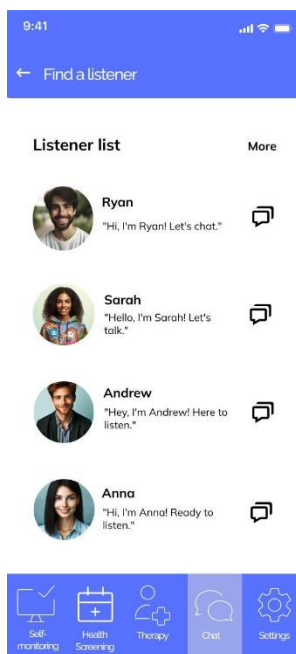


Figure 48: Find a listener page

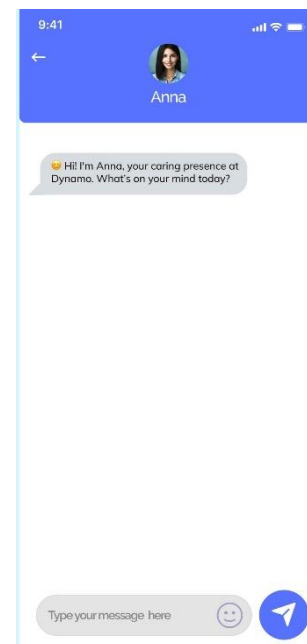
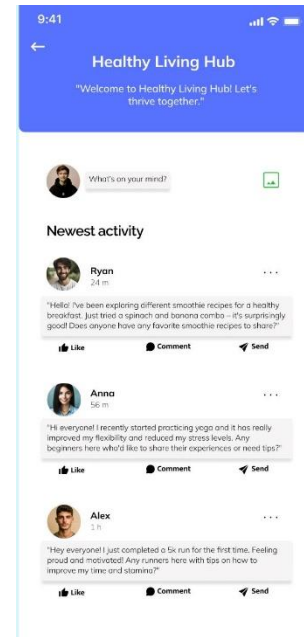


Figure 49: Chat with listener (Anna)



Figure 50: Join a community page



**Figure 51: Community page
(Healthy Living Hub)**

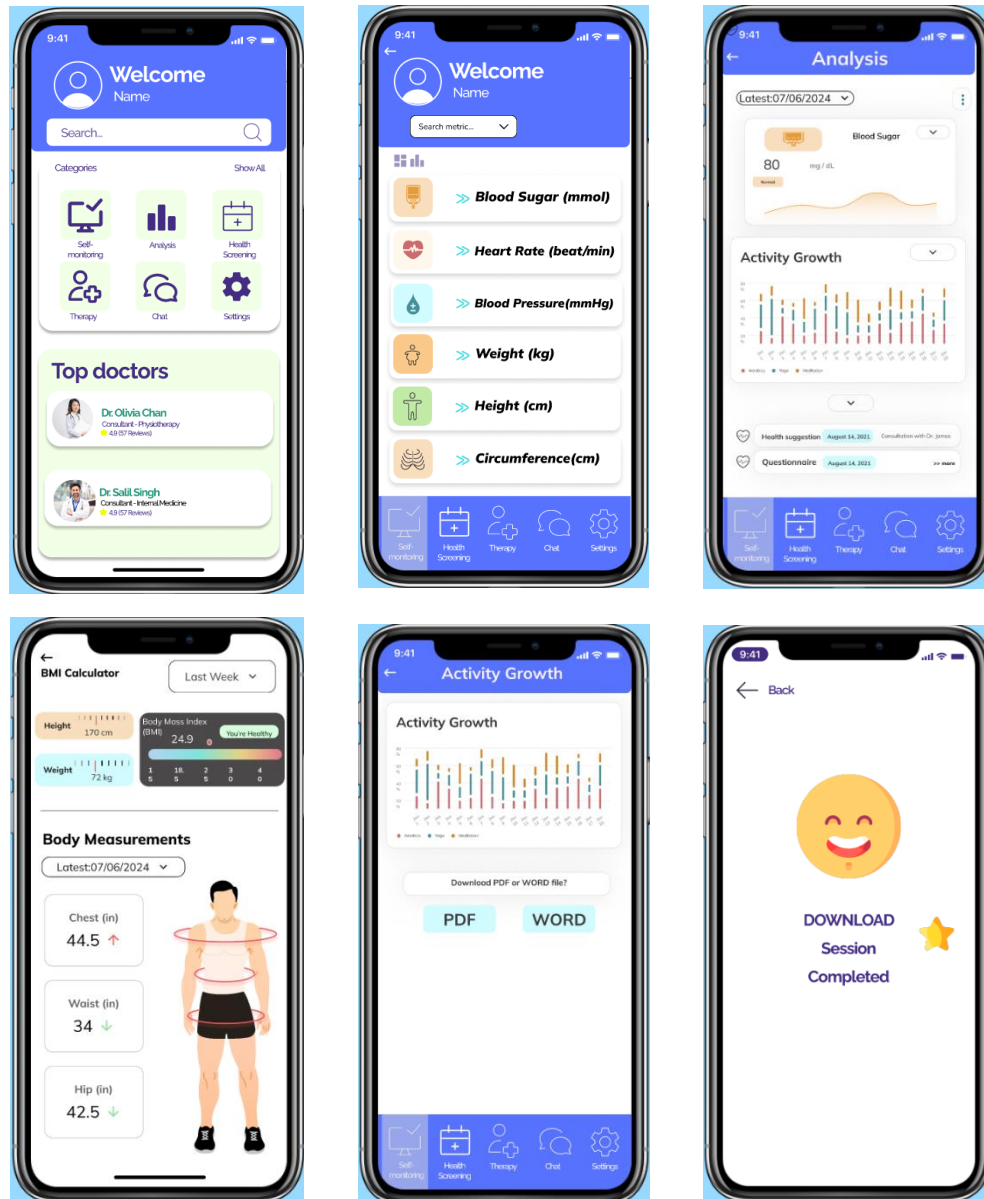
3.0 Briefing

Good morning Sir/Madam, today we are going to show you our application called Medico. Our app is a Health Care application which enables people with disabilities especially those who suffer from mental health issues, physical health issues and elderly people to lead healthier and more fulfilling lives.

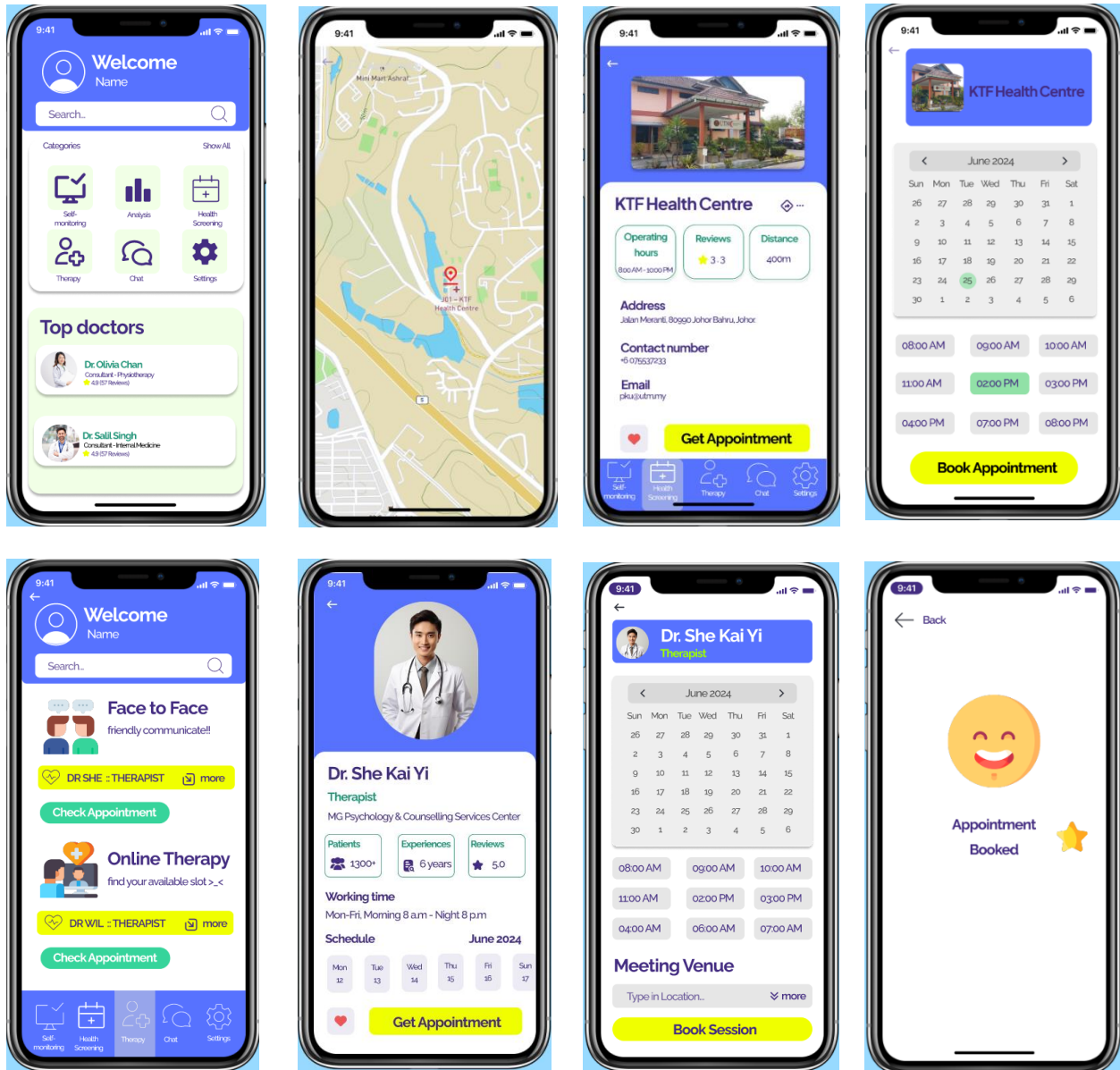
Sir/Madam, before we start, may I have your verbal consent first and the permission to record this session? If yes, I'll continue explaining the 3 tasks by using the prototype that we designed in Figma. First, you need to manage your healthcare data record and see the analysis of that data, then book a health screening appointment and therapy session, and lastly use the online chatting platform. If for whatever reason Sir/Madam is unable to complete the task or could not figure out how to complete the task, you can say "terminate". Remember to think aloud as you interact with the app. Describe what you are doing and why so we can understand your thought process. You can start whenever you are ready.

4.0 Testing with users

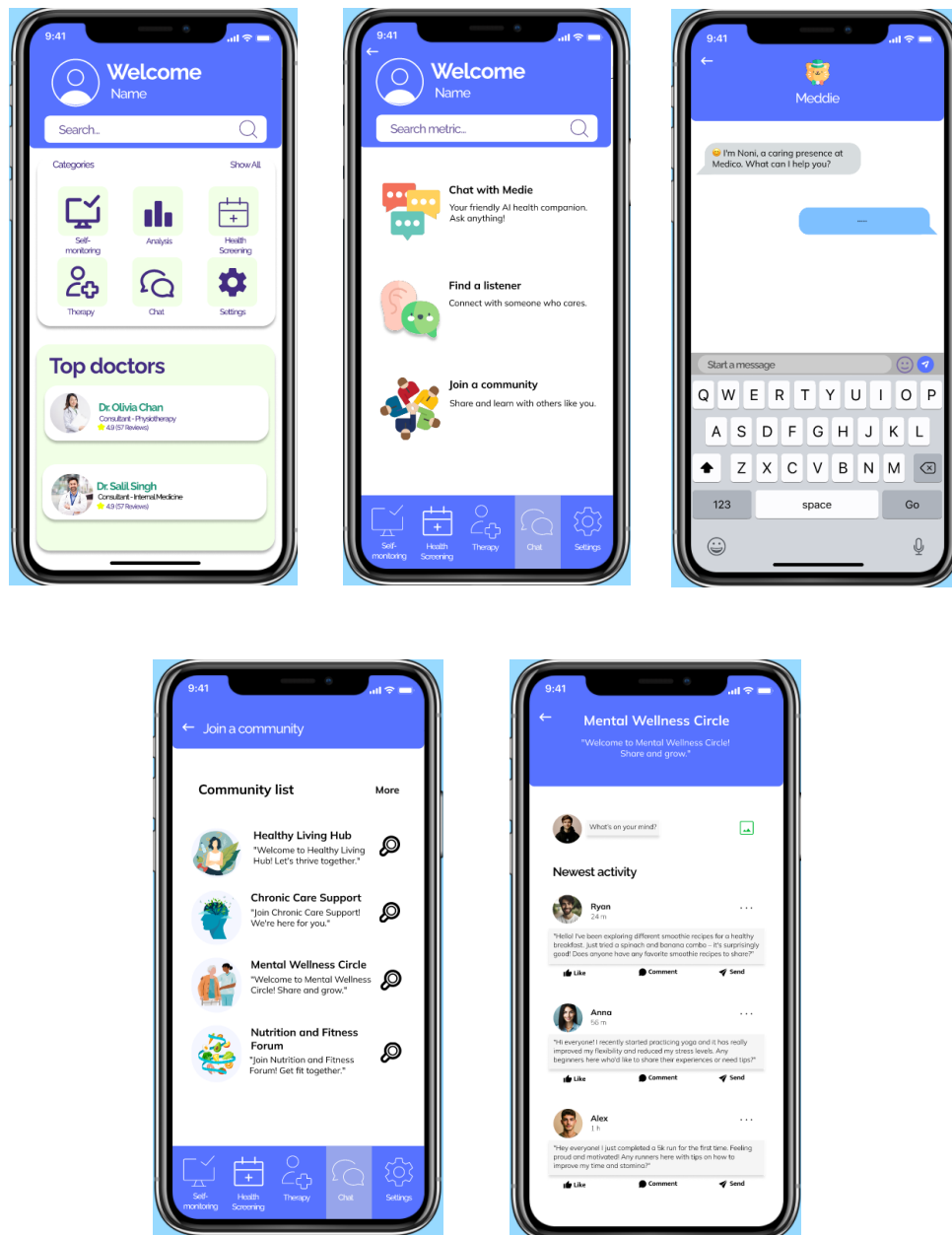
Task 1: Review the self-monitoring analysis



Task 2: Book health care services



Task 3: Explore the chatting platform



4.1 User 1:

Testing Video With User 1:

https://drive.google.com/drive/folders/1ywKkaqXHn8_0WNlkqw24gAms2GZXpzJ2

4.2 User 2:

Testing Video With User 2:

<https://drive.google.com/drive/folders/1vPt5zjXnZhX96gxEgWadImt9dDiRyYOS>

4.3 User 3:

Testing Video With User 3:

https://drive.google.com/file/d/15M81UOgKd-8_z6prs5mv57yI66Pzzalp/view?usp=drive_link

5.0 Observation

In order to test our health care app, Medico, we had found 3 potential users based on our target users, which are individual with physical health problem, elderly and individual with mental health problem. The testing was done in interview way, we showed our Figma prototype to the users and asked them to do three tasks based on our main functions, which include review the self-monitoring analysis, book health care services and explore the chatting platform.

Based on the feedback from user 1, the overall design of the app is good. The layout of the app is neat, as in the home page, he can see all of the button icon, and all of the button works properly. Moreover, the use of the icons and colours are appropriate, which make the app clean and user-friendly. For the functions part, he said that the functions of the app are comprehensive. For self-monitoring function, he can view all of his personal health with different date and also the graph analysis. For the health care services, he said that the functions are easy to use, while for the chatting platform, he said that the function can help user with different requirement because they can either chat with AI chatbot, find a listener or join a community.

Based on the feedback from User 2, the overall design of the app is highly praised for its neat and user-friendly layout. The app provides a smooth and straightforward login process, making it easy for users to access its features. The user can effortlessly track their personal health data, including blood sugar levels and monthly activity growth, through the self-monitoring function. Additionally, the app allows users to check their heart rate and other health metrics with monthly activity growth displayed in a clear, graphical format. Users can easily book free health screenings and find the nearest health clinics. However, the user didn't get to book the appointment for free doctor services which located in the doctor profile page, as the user didn't choose to open the profile page. The app is easy for the user to check appointments and communicate with doctors, either through chat or by entering meeting rooms. Furthermore, the user can easily access to the interactive chatting platform which offers multiple options for user support, including chatting with an AI friend, finding a listener, and joining a community, thus catering to diverse user needs. Overall, the app's design and functionality are tailored to be intuitive and efficient, enhancing the user experience by making health management and communication more accessible.

Based on the feedback from User 3, the overall design of the app is simple and easy to understand. All icons are directly linked to the relevant user interfaces, making it easier for the elderly and people with mental health issues to navigate the application. The user interface layout is both interactive and attractive, providing a comfortable experience for users. Additionally, when users attempt to complete Task 2, which involves making healthcare service appointments, the app helps those with mental health issues save time and offers a better platform to communicate and deal with hospital staff. The app allows users to book appointments efficiently, ensuring that their needs are verified and addressed by certified doctors. Finally, the chatting platform is straightforward. The AI communication, combined with having a verified listener and an interested community, enhances the overall user experience within the application.

6.0 Findings

User 1:

1: There is no booking button on the health screening page. Users may be confused about how to book an appointment for health screening.

Solution: Add a clear booking button to the health screening page to facilitate appointment scheduling.

2. There is no option to upload photos on the chat page, so users cannot send photos when chatting with other people.

Solution: Add an upload photo option to the chat page to enable users to share photos during their conversations.

User 2:

1. The system didn't provide the booking appointment button in the therapy page. This causes the user didn't get to book the appointment for free doctor services which located in the doctor profile page, as the user didn't choose to open the profile page.

Solution: The booking process could be improved by making the "book appointment" button more accessible outside of the doctor profiles.

2. The system is lack of clarity in the self-monitoring system, leading to difficulties in understanding the information presented. Users, especially those without a medical background, struggle to interpret their health metrics and prioritise which data points are critical. For example, user is unsure whether her blood sugar levels, or heart rate should be more closely monitored, indicating a need for better guidance within the app.

Solution: Redesign the data presentation format to be more intuitive and user-friendly. Use colour codes for example green for normal, yellow for caution, red for danger to help users quickly understand the status of their health metrics.

3. The health data analysis system also falls short in providing clear instructions on how to read and understand the analysis data. This lack of comprehensive explanations and user-friendly instructions can leave users feeling confused and unable to effectively utilise the health information provided by the app to make informed decisions about their health.

Solution: Integrate AI educational tooltips and tutorials within the app that explain the analysed health data. For instance, when a user views their blood sugar or heart rate data, a brief, accessible explanation of what the numbers mean and why they are important should be provided.

User 3:

1. User 3 found that the icons used in the app were not always clear or intuitive, especially for elderly users. This made it difficult for them to navigate the app efficiently.

Solution: Redesign the icons to be more universally recognizable and include text labels under each icon. This would help all users, particularly the elderly, to understand the functions of each icon more easily.

2. Difficulty in Understanding AI Chatbot Responses

Solution: Simplify the language used by the AI chatbot and provide more straightforward, user-friendly responses. Additionally, implement a feedback mechanism where users can rate the helpfulness of the responses and suggest improvements.